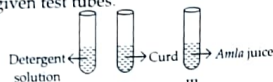
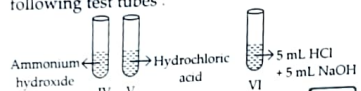


- (c) Distilled water, window cleaner and lime water
(d) Lime water, soap solution and lemon juice

Case II : Red rose indicator has been added to the given test tubes.



Phenolphthalein indicator has been added to following test tubes :



1. Colour change in the given test tubes will be

I	II	III
(a) Light pink	Red	Light green
(b) Green	Red	Red
(c) Magenta	Green	Green
(d) Green	Light pink	Red

2. The colour of solution in test tubes IV, V and VI will be
(a) pink, pink, colourless
(b) pink, colourless, colourless
(c) pink, pink, green
(d) colourless, pink, colourless.

3. Complete the given table by choosing correct option.

Test tube	Effect on blue litmus paper	Effect on turmeric paper
I	No change	Y
V	X	No change

- | | |
|---------------|--------|
| X | Y |
| (a) Red | Yellow |
| (b) Red | Blue |
| (c) Red | Red |
| (d) No change | Red |

4. How many of the given substances (I–VI) will turn red litmus to blue?

- | | |
|-------|-------|
| (a) 2 | (b) 3 |
| (c) 4 | (d) 6 |

*Acids → Hydrogen⁺ ions
Bases → Hydroxide⁻ ions*

11/6/20

The World of Metals and Non-metals

CHAPTER

02

Learning Outcomes...

- | | |
|---|--|
| <ul style="list-style-type: none"> Metals Physical and Chemical Properties of Metals and Uses Corrosion of Metals Prevention of Corrosion | <ul style="list-style-type: none"> Non-metals Physical and Chemical Properties of Non-metals and Uses Comparison of Physical and Chemical Properties of Metals and Non-metals |
|---|--|

Introduction

Elements have been broadly divided into two main groups depending on their properties, metals and non-metals. Besides these, there are some elements which show the properties of both metals and non-metals, they are called **metalloids**. All elements are arranged in the periodic table according to their properties. Presently 118 elements are known. These elements are the basic building blocks of all matter. Some are naturally occurring while others are artificially made in the laboratory, and do not exist in nature.

Metals

Metals are those elements which give electrons to form positive ions or cations, i.e., they are electropositive in nature. Generally they contain 1, 2, or 3 valence electrons. Few metals occur in free or native state, e.g., gold, silver and platinum. They are called **noble metals**. Reactive metals like sodium, aluminium, zinc, etc., occur in combined state in the form of chemical substances called **minerals**. The minerals from which metals can be extracted conveniently and profitably are called **ores**. The process of obtaining a pure metal from one of its ore is known as **metallurgy**.

Physical Properties of Metals

Metals can be distinguished from non-metals on the basis of their physical and chemical properties.

- Physical state :** Almost all metals are solids at room temperature (except mercury and gallium which are liquids) at room temperature.
- Melting and boiling points :** Melting and boiling points of metals are very high (except mercury and gallium).
- Hardness :** Metals are generally hard (except lithium, sodium and potassium which are soft metals and can be cut with a knife).
- Lustre :** Metals have a shine of their own and can also be polished. Lustre shown by metals is known as metallic lustre.
- Malleability and ductility :** The property of metals by which they can be beaten into thin sheets is called **malleability** and the property of metals by which they can be drawn into thin wires is called **ductility**. Metals possess these properties in varying proportions because their atoms are arranged in layers, so they can slip past each other when hammered or pulled. Gold and silver are the most malleable metals.

DO YOU KNOW?

Gold is so ductile that one gram of it can be drawn into a 2-kilometre-long wire approximately.

ACTIVITY CORNER

To show metals are malleable in nature

» **Materials required** : Iron nail, coal piece, pencil lead, aluminium wire and a hammer.

» **Procedure** : Take all the materials one by one and beat them with the help of a hammer repeatedly and note the change in shape.

» **Observations** : Iron nail and aluminium wire flatten while coal piece and pencil lead break into small pieces and become powder.

» **Conclusion** : Iron nail and aluminium wire are metals, hence they are malleable while coal piece and pencil lead are non-metals hence they are brittle.

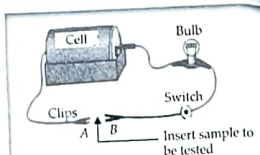
» **Conductivity** : Metals are good conductors of heat and electricity. Some metals allow electricity to pass through them more easily than the other metals. For example, copper and aluminium allow current to pass through them more easily as compared to iron or lead. Silver is the best conductor of heat and electricity.

ACTIVITY CORNER

To show that metals are good conductors of electricity

» **Materials required** : Cell, conducting wire, tester bulb, iron nail, coal piece, copper wire and sulphur.

» **Procedure** : Make an electric circuit with the help of a cell, wire and a bulb. Join the wires with the help of iron nail. Note the change in the bulb. Repeat the experiment by putting other objects in place of iron nail.



» **Observations** :

Iron nail – The bulb glows. Coal piece – The bulb does not glow. Copper wire – The bulb glows. Sulphur – The bulb does not glow.

» **Conclusion** : Iron nail and copper wire are metals, hence good conductors of electricity while coal piece and sulphur are non-metals and hence bad conductors of electricity.

» **Tensile strength** : Metals have high tensile strength. They can be stretched to a great extent without breaking.

» **Sonority** : The property of metals by which they produce ringing sound when struck is called sonority. Metals produce a ringing sound when struck hard, they are said to be sonorous.

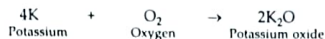
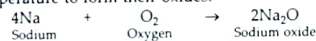
» **Density** : Metals generally have high densities as in metals atoms are closely packed (except lithium, sodium and potassium which have low densities).

Chemical Properties of Metals

Most metals react chemically with other substances in one form or another. Few metals which do not react with common chemicals are called **noble metals**, e.g., silver, gold, platinum. The rate of reaction for similar reactions is not same for all the metals. The rate of reaction depends on the reactivity of metals.

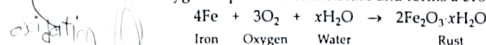
» **Reaction with oxygen** : Metals react with oxygen to form oxides, few metals react vigorously while few reacts only on heating. Examples :

– Sodium and potassium are most reactive metals. They combine with oxygen even at room temperature to form their oxides.



In fact, the reaction is so vigorous that these metals catch fire if kept in the open air. That is why they are always kept under kerosene oil.

– Iron reacts with oxygen in presence of moisture and forms a brown coloured powder called **rust**.

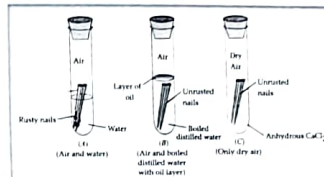


ACTIVITY CORNER

To show that rusting of iron requires both air and water

» **Procedure** : Take three glass tubes marked A, B and C provided with rubber corks.

Place a few clean iron nails in the tube marked 'A' pour some water. In the tube 'B' pour some boiled distilled water along with some oil say turpentine oil. Similarly, in the tube 'C' add some anhydrous calcium chloride or silica gel. Cork these tubes properly and leave them undisturbed for few days.

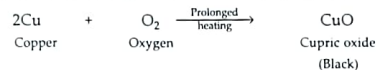


» **Observations** : The nails placed in the tube 'A' have been rusted while they remain unaffected in the tubes 'B' and 'C'.

In the tube 'A', the nails are exposed to both air and water and have been rusted. In the tube 'B', the oily layer above water does not allow air to come in contact with the nails. In the absence of air, these are not rusted. Anhydrous calcium chloride or silica gel placed in the tube 'C' absorbs the entire moisture or water vapours. Although the nails are in contact with air but still they do not get rusted.

» **Conclusion** : Both air and water are required for rusting of iron.

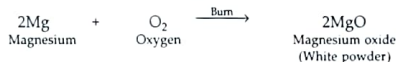
– Copper does not burn in air even on strong heating. However, on heating in air for a long time, it is covered with a layer of black coloured copper(II) oxide, i.e., cupric oxide.



– Magnesium, aluminium, zinc and lead combine with oxygen at ordinary temperatures to form a protective layer of the oxide on their surface. However, the reaction is much slower than that of sodium and potassium.



Further, on heating magnesium to ignition temperature, it burns with a dazzling white light to form magnesium oxide (MgO) which is a white powder.



This shows that magnesium is more reactive than aluminium, zinc and lead. Even among the latter three metals, the order of reactivity is $Al > Zn > Pb$ (Reaction between zinc and oxygen takes place only on heating)

Metal oxides are basic in nature. When these basic oxides are dissolved in water, they form bases. This can be tested by the help of a red litmus paper, which turns blue.

ACTIVITY CORNER

To check the nature of magnesium oxide formed by reaction of burning magnesium ribbon in air

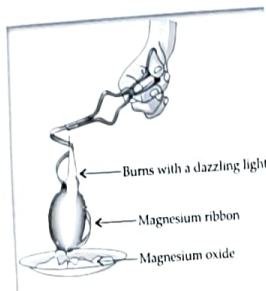
» **Materials required :** Magnesium ribbon, red

and blue litmus papers and burner.

» **Procedure :** Burn magnesium ribbon in air and collect the ash. Dissolve the ash in water and test with red and blue litmus papers.

» **Observations :** Red litmus paper turns blue.

» **Conclusion :** Magnesium oxide formed by burning magnesium with oxygen is basic in nature.



ACTIVITY CORNER

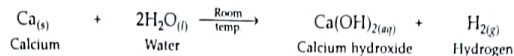
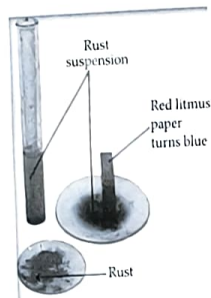
To check the nature of rust formed as a result of the reaction between iron, oxygen and water

» **Materials required :** Rust, test tube, water, watch glass, blue and red litmus papers.

» **Procedure :** Collect a spoonful of rust and dissolve it in a little amount of water, shake the suspension and test with red and blue litmus papers.

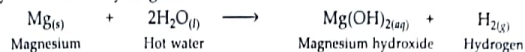
» **Observations :** Red litmus paper turns blue.

» **Conclusion :** Iron oxide (rust) is basic in nature.

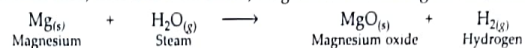


Further, the bubbles of hydrogen gas produced stick to the surface of calcium and hence it starts floating on the surface of water.

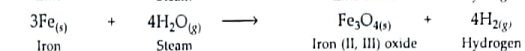
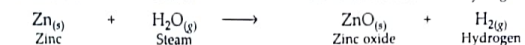
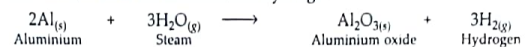
- Magnesium does not react with cold water. It reacts only with hot water forming magnesium hydroxide and hydrogen.



It also starts floating on the surface of water due to bubbles of hydrogen gas sticking on its surface. Also, note that with steam, magnesium forms magnesium oxide and hydrogen.



- Aluminium, zinc and iron neither react with cold water nor with hot water. They react only with steam to form metal oxide and hydrogen.



- Lead, copper, silver and gold do not react even with steam.

Thus, the order of reactivity of different metals towards water may be written as:

$K > Na > Ca > Mg > Al > Zn > Fe > Pb > Cu > Ag > Au$

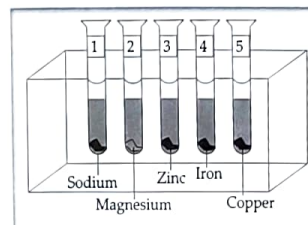
ACTIVITY CORNER

To compare the reactivity of some metals with water and arrange them in order of their increasing reactivity

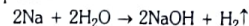
» **Materials required :** Small

pieces of metals like sodium, magnesium, zinc, iron and copper, five test tubes, burner and test tube holder.

» **Procedure :** Put a small piece of each metal in different test tubes numbered 1 to 5. Add 5 mL of water in each test tube.



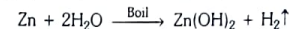
» **Observations :** In test tube-1, in which sodium metal was taken, reaction takes place vigorously. A gas is evolved which burns with a pop sound.



In test tube-2, in which Mg was taken, there is no reaction in cold water. When the test tube is heated, magnesium starts reacting and bubbles of hydrogen gas ~~can~~ be seen escaping from the test tube.



In test tube-3, in which Zn was taken, reaction takes place only on boiling the water.



In test tube-4, in which iron is taken, there is no reaction even when the water is boiled. Fe reacts slowly when it is heated with steam. In test tube-5, which has copper and water, no reaction takes place even with steam.

» **Conclusion :** The reactivity order of the metals in the reaction with water is :

$Na > Mg > Zn > Fe > Cu$

As we have seen by the reactions of metals with oxygen and water, different metals react with these substances with different rates. The rate of reaction depends on the reactivity of various metals.

On the basis of their reactivity, a series known as **reactivity series** also known as activity series has been drawn in which metals are arranged in decreasing order of their reactivity. A metal can replace another metal placed below it in the reactivity series, but cannot replace a metal placed above it in the reactivity series.

Metal reactivity series :

	Potassium	K → Most Reactive Metal	
	Sodium	Na	
	Calcium	Ca	
	Magnesium	Mg	
	Aluminium	Al	
	Zinc	Zn	
	Iron	Fe	
	Nickel	Ni	
	Tin	Sn	
	Lead	Pb	
			↓ REACTIVITY DECREASES
	Hydrogen	H	
	Copper	Cu	
	Mercury	Hg	
	Silver	Ag	
	Gold	Au → Least Reactive Metal	

Metals more reactive than hydrogen

Metals less reactive than hydrogen

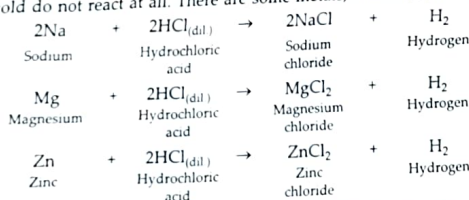
DO YOU KNOW?

▶ The most expensive metal is rhodium, however, platinum and gold, numbers two and three respectively.

▶ Aluminium is widely used in making beverage cans and recycling one aluminium can saves enough energy to power a television for approximately three hours. This is because recycling aluminium requires about 95% less energy than producing it from raw materials.

COMPETITION WINDOW

- **Reaction with acids :** Metals react with dilute acids like hydrochloric acid and sulphuric acid to form corresponding salts and liberate hydrogen. The reaction becomes less vigorous as we move down the reactivity series. Sodium and potassium mostly show violent reaction, whereas magnesium and aluminium react quite vigorously. Iron and zinc show moderate reaction, while tin and lead react slowly. Silver and gold do not react at all. There are some metals, which react only with concentrated acids.



ACTIVITY CORNER

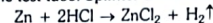


To show the reaction of metals with acids

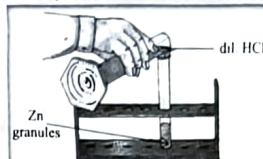
▶ **Materials required :** Zinc granules, dilute hydrochloric acid and burning splinter.

▶ **Procedure :** Take some zinc granules in a test tube. Add little dilute hydrochloric acid to it.

▶ **Observations :** A gas is seen coming out of the test tube in the form of bubbles. Bring a burning splinter near the mouth of the test tube. Splinter burns with a pop sound.



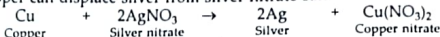
▶ **Conclusion :** Metals react with acids evolving hydrogen gas.



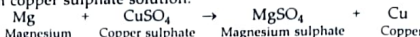
COMPETITION WINDOW

- **Reaction with salt solutions :** Metals react with salt solutions of different metals. More reactive metals displace less reactive metals from their salt solutions. Such reactions are called **displacement reactions**. Examples :

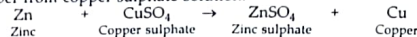
- Copper can displace silver from silver nitrate salt.



- When a ribbon of magnesium is dipped in copper sulphate solution, it displaces copper from copper sulphate solution.



- When zinc metal is dipped in copper sulphate solution, zinc being more reactive displaces copper from copper sulphate solution.



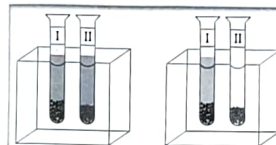
ACTIVITY CORNER



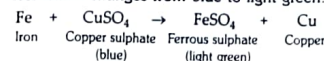
To show that more reactive metals can displace less reactive metals from their salt solutions

▶ **Materials required :** Iron filings, zinc metal, ferrous sulphate solution, copper sulphate solution, test tubes and stand.

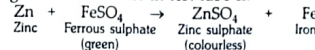
▶ **Procedure :** Take two test tubes. Put about 5 mL of copper sulphate solution in test tube-I, and 5 mL of ferrous sulphate solution in test tube-II. Place some iron filings in test tube-I and a piece of zinc metal in test tube-II. Allow the test tubes to stand for sometime.



▶ **Observations :** The colour of the solution in test tube-I changes from blue to light green.



The colour of the solution changes from light green to colourless in test tube-II.



▶ **Conclusion :** Since iron is more reactive than copper, it displaces copper from copper sulphate solution and blue colour of copper sulphate disappears due to the formation of ferrous sulphate which is light green.

Zinc is more reactive than iron, therefore it displaces iron from ferrous sulphate solution and light green colour of ferrous sulphate gradually becomes colourless, due to the formation of zinc sulphate. Hence, more reactive metals can displace less reactive metals from their salt solutions.

ILLUSTRATIONS

1 Identify the metal on the basis of the property mentioned.

- Liquid at room temperature.
- Develops a green layer on the surface, in presence of moist air.
- It is stored in kerosene.
- Has magnetic properties and is rusted.

Ans.: (a) Mercury (b) Copper (c) Sodium (d) Iron

2 Complete the following reactions :

- $\text{Mg} + \text{HCl} \rightarrow \dots + \text{H}_2$
- $\text{Ca} + \dots \rightarrow \text{CaSO}_4 + \dots$
- $\dots + \text{HCl} \rightarrow \text{ZnCl}_2 + \dots$
- $\dots \rightarrow \text{MgSO}_4 + \text{H}_2$
- $\text{Al} + \text{ZnSO}_4 \rightarrow \dots + \dots$

Ans.: (i) $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$

(ii) $\text{Ca} + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2$

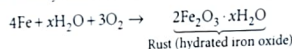
Uses of Metals

- Metals are used to make electric wires, since they are good conductors of electricity. Examples include copper and aluminium.
- Some metals such as zinc are used for galvanizing iron, as a protective layer against rusting.
- Metals such as aluminium, copper and iron are used to make utensils.
- Mercury is used in thermometers.
- Metals such as silver, gold and platinum are used to make jewellery.
- Nickel and chromium are used for electroplating.
- Metals, being very hard and strong, are used as building materials.

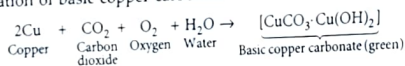
Corrosion of Metals

If a metal is reactive, its surface is attacked slowly by air and water (moisture). There is a chemical reaction between the metal and moist air, which tarnishes the metal, e.g., iron, copper, silver etc. Gradually, the chemical reaction takes place underneath the surface as well and the whole metal is eaten away.

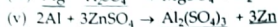
- Iron** : Iron when exposed to moist air forms a brown powdery substance called rust. It is not confined to the surface and slowly the whole metal is eaten up.



- Copper** : Copper is quite unreactive metal. However, it is also corroded, though slowly, when it is exposed to moist air. When a copper object is exposed to moist air, it forms a greenish layer which is due to the formation of basic copper carbonate.



- Silver** : When silver objects are exposed to air, they are tarnished, i.e., they lose their shine. Silver is quite an unreactive metal, therefore it does not react easily with oxygen. However, it combines with sulphur which is present in the atmosphere, to form a black coating of silver sulphide (Ag_2S).



3 Why is sodium metal stored in kerosene?

Ans.: Sodium metal is very reactive. It reacts vigorously with oxygen and water. A lot of heat is generated in the reaction. It is therefore, stored in kerosene.

4 What are noble metals? Why are they called so?

Ans.: The noble metals are metallic elements like silver, gold, platinum, iridium, palladium. These metals are called noble metals because they are non-reactive and occur in free state in nature. They maintain their metallic lustre for a long time and do not get corroded. They remain unaffected by atmospheric air, moisture or common chemicals like acids and bases.

- Aluminium** : Aluminium vessels lose their shine after sometime due to formation of a layer of aluminium oxide on the surface of the metal. This thin layer of oxide is unreactive and protects the metal from further damage due to corrosion.

Prevention of Corrosion

Iron is most easily protected by paint which provides a barrier between the metal and air or water. Moving parts of machines can be protected by a water repellent oil or grease layer. Covering the surface with enamel and lacquer is another method.

Few other methods to prevent rusting like galvanization, electroplating and alloying are discussed below:

Galvanizing

Coating iron or steel with a thin zinc layer is called 'galvanizing'. This layer is produced by electrolytic deposition. Dipping the iron or steel object in molten zinc and using it as the negative electrode zinc is coated on iron or steel used as cathode. Zinc preferentially corrodes or oxidizes to form a zinc oxide layer that does not flake off like rust. Also, if the surface is scratched, the exposed zinc being more reactive again corrodes in preference to iron and continues to protect it.

Electroplating

The process of coating a less reactive metal with a more reactive metal by electrolysis is known as electroplating.

For electroplating	Anode	Cathode	Electrolyte
With copper	Cu	Object	CuSO_4 + dilute H_2SO_4
With silver	Ag	Object	$\text{K}[\text{Ag}(\text{CN})_2]$
With nickel	Ni	Object	Nickel ammonium sulphate
With gold	Au	Object	$\text{K}[\text{Au}(\text{CN})_2]$
With zinc	Zn	Iron objects	ZnSO_4
With tin	Sn	Iron objects	SnSO_4

Alloying with Nickel and Chromium

When iron is alloyed with chromium, nickel and some carbon [$\text{Fe} = 73\%$, $\text{Cr} = 18\%$, $\text{Ni} = 8\%$], stainless steel is obtained. Stainless steel resists corrosion and hence does not rust. For example, cooking utensils, cutlery, ornamental pieces and surgical instruments are made up of stainless steel and hence do not rust at all.

Improving Properties of Metals by Formation of Alloys

Certain properties of the metals such as lustre, malleability, ductility, resistance to action by air and moisture etc. can be improved by the addition of small amounts of other metals or even non-metals. For example, pure gold (also called 24 carat gold) is very soft. It cannot be used for making jewellery. A small amount of silver or copper is mixed with it in order to improve its hardness, malleability and ductility. This is normally 22 carat gold and is used in making ornaments. The homogeneous mixture of more than one metal or even metal and non-metal is known as alloy. It is prepared by melting the main constituting metal and then adding definite proportions of the other elements (metals or non-metals) also in the molten state. The molten mass is uniformly stirred and cooled to form a solid mass. This is known as alloy. The alloy in which mercury is one of the constituents is known as amalgam.

COMPETITION WINDOW

Alloys, Their Constituents and Uses

COMPETITION WINDOW

S.No.	Alloy	Constituents	Uses
1	Brass	Cu and Zn	Harder than pure Cu and Zn; used for making utensils, cartridges
2	Bronze	Cu and Sn	For making statues, medals, ships, coins, machines, etc.
3	Solder	Sn and Pb	For joining metals, soldering wires, electronic components, etc.
4	Duralumin	Al, Cu and traces of Mn and Mg	In bodies of aircrafts, kitchen ware, automobile parts, etc.
5	German silver	Cu, Zn and Ni	For making utensils, ornaments.
6	Gun metal	Cu and Sn	Gears, castings, etc.
7	Bell metal	Cu and Sn	Bells, gongs, etc.
8	Magnalium	Al and Mg	Balance beams, light instruments.
9	Type metal	Pb, Sb and Sn	Casting type
10	Stainless steel	Fe, Cr, Ni and traces of carbon	Utensils, cutlery etc.

Non-metals

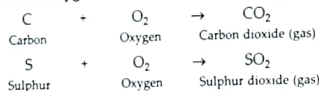
Non-metals are those elements which accept electrons to form negative ions or anions, i.e., they are electronegative in nature. Generally they contain 5, 6 or 7 valence electrons. Most of the non-metals are used in daily life in the form of compounds or in various other forms.

Physical Properties of Non-metals

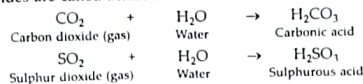
- Non-metals exist in all three states; solid, liquid and gas, e.g., carbon, sulphur, phosphorus are solids, bromine is a liquid, hydrogen, nitrogen and chlorine are gases.
- Most of the non-metals are soft except diamond which is the hardest material known.
- Solid non-metals are soft and dull. They break down into a powdery mass on tapping with a hammer. For example, coal and sulphur.
- Non-metals are not sonorous.
- They are poor conductors of heat and electricity except graphite which is a good conductor of electricity.
- They do not possess metallic lustre except solid iodine.
- They are neither malleable nor ductile.

Chemical Properties of Non-metals

- Reactions with oxygen : Non-metals react with oxygen on heating and form their respective oxides.



These oxides when dissolved in water, form acids which turn blue litmus red. Therefore, these oxides are called acidic oxides.



ACTIVITY CORNER

To show that non-metal oxides are acidic in nature

- » **Materials required :**
Powdered sulphur, deflagrating spoon,

water, red and blue litmus paper, gas jar.

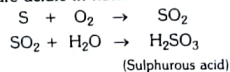
- » **Procedure :** Take a small amount of sulphur in a deflagrating spoon and heat it. Put the spoon in a gas jar or a glass tumbler with a lid as soon as sulphur starts burning. Remove the spoon after some time and add some water into the jar and shake well. Check the solution with red and blue litmus paper.

- » **Observations :** Blue litmus paper turns red.

Generally, non-metals do not react with water or acids, since they cannot displace hydrogen as metals do. Few non-metals are very reactive in air, hence stored in water e.g., phosphorus.

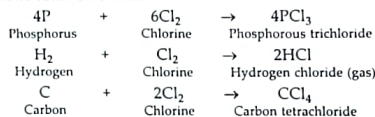


- » **Conclusion :** Sulphur reacts with oxygen to form sulphur dioxide which when dissolved in water forms sulphurous acid which turns blue litmus red. It shows that non-metal oxides are acidic in nature.

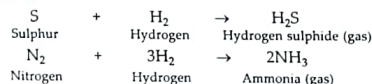


COMPETITION WINDOW

- Reactions with chlorine : Non-metals react with chlorine to form chlorides.



- Reactions with hydrogen : Non-metals combine with hydrogen and form corresponding compounds.



DO YOU KNOW?

- » Coming up too fast from deep sea diving gives the "bends" which is nitrogen bubbles in your blood.
- » The sound you hear when you crack your knuckles is actually nitrogen gas bubbles popping.

Uses of Non-metals

- Oxygen is essential for all living beings for breathing.
- Hydrogen is used in welding metals.
- Sulphur is used in the manufacture of sulphuric acid and it is used to make antiseptic creams. Sulphur is also used in the vulcanization of rubber.
- Liquid hydrogen is used as a rocket fuel.
- Nitrogen is used in the manufacture of ammonia and fertilizers. It is also used as preservative due to its inert nature. Some compounds of nitrogen are used as explosives as well.
- Phosphorus is used in safety materials and in fire crackers.
- Graphite, a form of carbon, is used in pencil leads and as electrodes. Diamond, another form of carbon is the hardest substance known, hence, used to cut glass.
- Silicon is used to make computer chips, silica gel is used to absorb the moisture.
- Chlorine is used in treatment of water to kill germs.

Comparison of Physical Properties of Metals and Non-metals

S. No.	Property	Metals	Non-metals
1	Lustre	Metals have lustre and can also be polished.	Except iodine, non-metals do not have lustre and cannot be polished.
2	Hardness	Metals (except sodium and potassium) are generally hard.	Non-metals (except diamond and boron) are generally soft.
3	Malleability and ductility	Metals are malleable (can be hammered into thin sheets) and ductile (can be drawn into wires).	Non-metals are neither malleable nor ductile. They are actually brittle.
4	Conductivity	Metals are good conductors of heat and electricity.	Non-metals (except graphite) are bad conductors of heat and electricity.
5	Melting points and boiling points	Metals (except mercury and gallium) generally have high melting and boiling points.	Non-metals (except carbon and boron) have low melting and boiling points.
6	Sonorosity	Metals are sonorous (<i>i.e.</i> , produce sound when hit with a hard object).	Non-metals are non-sonorous (<i>i.e.</i> , do not produce any sound when hit with a hard object).
7	Density	Metals generally have high density.	Non-metals generally have low density.
8	Tensile strength	Metals generally have high tensile strength and hence cannot be easily broken.	Non-metals, generally, have low tensile strength and hence can be easily broken.
9	Physical state	Metals are generally solids (except mercury and gallium which are liquids) at room temperature.	Non-metals exist in all three states of matter.

Comparison of Chemical Properties of Metals and Non-metals

Property	Metals	Non-metals
1. Nature of ions	Metals are electropositive elements and hence lose one or more electrons to form positive ions. For example, $\text{Na} \rightarrow \text{Na}^+ + e^-$ Sodium Sodium ion $\text{Ca} \rightarrow \text{Ca}^{2+} + 2e^-$ Calcium Calcium ion	Non-metals are electronegative elements and hence gain one or more electrons to form negative ions. For example, $\text{Cl} + e^- \rightarrow \text{Cl}^-$ Chlorine Chloride ion $\text{O} + 2e^- \rightarrow \text{O}^{2-}$ Oxygen Oxide ion

2	Nature of oxides	Metals form basic oxides. For example, oxides of Na and Mg dissolve in water to form bases which turn red litmus blue. $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ $\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH (Base)}$	Non-metals form acidic oxides or neutral oxides. For example, oxides of nitrogen and sulphur dissolve in water to form acids which turn blue litmus red. $4\text{NO}_2 + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HNO}_3 \text{ (Acid)}$ CO, NO and H ₂ O are neutral
3	Reaction with water/steam	Some active metals like Na, K, Ca, Al, Fe, etc. displace hydrogen from water or steam.	Non-metals (except fluorine) generally do not react with water. $2\text{F}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HF} + \text{O}_2$ Hydrofluoric acid Oxygen
4	Reaction with dilute acids	Metals which lie above hydrogen in the activity series displace hydrogen from dilute acids. For example, $\text{Zn} + \text{dil. H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2 \uparrow$	Non-metals do not react with dilute acids and hence do not displace hydrogen from dilute acids.
5	Oxidising and reducing behaviour	Metals have strong tendency to lose electrons and hence they behave as reducing agents. For example, Na reduces Cl ₂ to Cl ⁻ ions. $2\text{Na} \rightarrow 2\text{Na}^+ + 2e^-$ $\text{Cl}_2 + 2e^- \xrightarrow{\text{Reduction}} 2\text{Cl}^-$ Chlorine Chloride ion	Non-metals have a strong tendency to accept electrons and hence they behave as oxidising agents. For example, Cl ₂ oxidises Na to Na ⁺ ions. $2\text{Na} \xrightarrow{\text{Oxidation}} 2\text{Na}^+ + 2e^-$ Sodium ion $\text{Cl}_2 + 2e^- \rightarrow 2\text{Cl}^-$

ILLUSTRATIONS

- 5** What are the main differences between metals and non-metals?

Ans.:

Metals	Non-metals
Mostly solid at room temperature (except Hg)	May be solid, liquid or gas at room temperature
Metallic lustre	Usually dull (except iodine and graphite)
Dense	Less dense
Malleable	Non-malleable
Ductile	Non-ductile
Sonorous	Non-sonorous
Hard and strong	Soft (except diamond)
Good conductors of heat and electricity	Poor conductors of heat and electricity (except graphite)
Form basic oxides	Form acidic oxides

- 6** Write some important uses of phosphorus and sulphur.

Ans.: (i) Phosphorus is used in manufacture of phosphoric acid and phosphate fertilizers.
 (ii) Phosphorus is used in the preparation of fire works, smoke screens and in matchstick industry.
 (iii) Sulphur is used in making sulphuric acid which in turn is used as a very important chemical reagent.
 (iv) Sulphur is used in vulcanization of rubber.
 (v) Sulphur is used as an antiseptic in skin ointments and as a germicide.

- 7** Write various methods to prevent corrosion.

Ans.: (i) Galvanization : Coating the surface of the metal with a non-corrosive metal like zinc is called galvanization.
 (ii) Electroplating : Plating of metal surface by less corrosive metal by using electricity is called electroplating. Chromium and nickel are used for electroplating.
 (iii) Painting the surface of the metal.

8 Give names of the following metals :

- The best conductor of electricity
- The metal used for preventing rusting of iron
- A metal which can be cut with a knife
- A metal which is present in the blood

Ans.: (i) Silver (ii) Zinc (iii) Sodium (iv) Iron.

9 Name the non-metals which show the following properties:

- A non-metal which has lustre and is a good conductor of electricity.
- A non-metal which is the hardest natural substance.
- A non-metal which is essential for respiration.
- A non-metal which is the source of energy in the sun and stars.

Ans.: (i) Graphite (carbon) (ii) Diamond (carbon)
(iii) Oxygen (iv) Hydrogen

10 (a) Define corrosion.

(b) What is corrosion of iron called?

(c) How will you recognise the corrosion of silver?

(d) Why corrosion of iron is a serious problem?

(e) How can we prevent corrosion of iron?

Ans.: (a) The process of slowly eating up of metals due to their conversion into oxides, carbonates, sulphides, etc., by the action of atmospheric gases and moisture is called corrosion.

(b) The corrosion of iron is called rusting.

(c) Silver articles become black after sometime when exposed to air. This is due to formation of a coating of black silver sulphide (Ag_2S) on its surface by the action of H_2S gas present in the air.

(d) Corrosion of iron is a serious problem. Every year large amount of money is spent to replace damaged iron articles. Corrosion causes damage to car bodies, bridges, iron railings, ships and to all objects made of iron.

(e) Corrosion of iron can be prevented by painting, greasing, galvanising, anodising, electroplating or making alloys.

CONCEPT MAP

THE WORLD OF METALS AND NON-METALS

Depending upon the physical and chemical properties materials are divided into different types.

Metals Those elements which lose electrons to form positive ions.

Physical properties

- Physical state : hard solids
- Very high melting and boiling points
- These can be beaten into thin sheets, i.e., these are malleable.
- These can be drawn into wires i.e., these are ductile.
- Good conductors of heat and electricity.
- They have lustre.

Chemical properties

- Reaction with oxygen : Metals form basic oxides except noble metals.
- Few metals react vigorously with water.
- Few metals react with acids and release hydrogen.
- Can displace less reactive metals from their salt solution.

Uses

- In making electrical wires e.g., copper and aluminium.
- In making utensils e.g., aluminium
- For making jewellery e.g., silver, gold, platinum, etc.

Corrosion of Metals

- If a metal is reactive, its surface is attacked slowly by air and water (moisture). There is a chemical reaction between the metal and moist air, which tarnishes the metal, e.g., iron, copper, silver.

Rusting of iron can be prevented by :

- Painting
- Applying grease and oil
- Electroplating iron by a more reactive metal like tin or chromium
- Galvanization of iron by zinc or tin

Non-metals Those elements which accept electrons to form negative ions.

Physical properties

- Physical state : solid, liquid or gas
- Neither malleable nor ductile
- Poor conductors of electricity and heat.
- Do not have lustre.

Chemical properties

- Reaction with oxygen : Non-metals form acidic oxides.
- Generally do not react with water.
- Do not react with acids.

Uses

- Non-metals are essential for our life.
- Used in fertilizers to enhance growth of plants.
- Used in water purification.
- Used in crackers.

Metalloids These are the elements which are less reactive than metals and more reactive than non-metals. These have intermediate properties.

Let us Enhance Our Learning

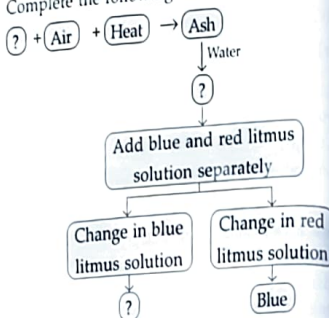
- Which metal is commonly used to make food packaging materials as it is cheaper, and its thin sheets can be folded easily into any shape?
 - Aluminium
 - Copper
 - Iron
 - Gold
- Which of the following metal catches fire when it comes in contact with water?
 - Copper
 - Aluminium
 - Zinc
 - Sodium
- State with reason(s) whether the following statements are True [T] or False [F].
 - Aluminium and copper are examples of non-metals used for making utensils and statues. []
 - Metals form oxides when combined with oxygen, the solution of which turns blue litmus paper to red. []
 - Oxygen is a non-metal essential for respiration. []
 - Copper vessels are used for boiling water because they are good conductors of electricity. []
- Why are only a few metals suitable for making jewellery?
- Match the uses of metals and non-metals given in Column-I with the jumbled names of metals and non-metals given in Column-II.

Column-I	Column-II
(i) Used in electrical wiring	(a) E N X Y G O
(ii) Most malleable and ductile	(b) N E C O H I R L
(iii) Living organisms cannot survive without it.	(c) P E P O R C

(iv) Plants grow healthy when fertilisers containing it are added to the soil.	(d) T E N G O I N R
(v) Used in water purification	(e) O G D L

- What happens when oxygen reacts with magnesium and sulphur. What are the difference in the nature of products formed?

- Complete the following flow chart:



- You are provided with the following materials. Discuss which material would be your choice to make a pan that is most suitable for boiling water and why?
Iron, copper, sulphur, coal, plastic, wood, cardboard
- You are provided with three iron nails, each dipped in oil, water and vinegar. Which iron nail will not rust, and why?
- How do the different properties of metals and non-metals determine their uses in everyday life?
- One of the methods of protecting iron from getting rusted is to put a thin coating of zinc metal over it. Since sulphur does not react with water, can it be used for this purpose? Justify your answer.
- An ironsmith heats iron before making tools. Why is heating necessary in this process?

Multiple Choice Questions

LEVEL - 1

Metals and their Physical and Chemical Properties

- When a substance X is hit with a hammer, it expands in size but does not break. This is because the substance is
 - ductile
 - hard
 - elastic
 - malleable.

(NCERT Page 42) **U**
- Which of the following is not a property of aluminium?
 - It is a good conductor of heat and electricity.
 - It is malleable.
 - It is heavy.
 - It is ductile.

R
- Mercury is used in thermometers because
 - it does not stick to the glass
 - it expands on heating
 - it is a liquid
 - all of these.

U (Page 43)
- Copper sulphate solution can be safely kept in a container made of
 - aluminium
 - lead
 - silver
 - zinc.
- Which of the following metals occurs in free state in nature?
 - Aluminium
 - Calcium
 - Gold
 - Sodium
- Aluminium is a type of metal that is used to make
 - wires and bulb filaments
 - tanks and magnets
 - drink cans, foils and aeroplanes
 - food containers and tooth fillings.
- The reason(s) why aluminium is preferred over copper for making overhead transmission cables
 - aluminium is lighter than copper
 - aluminium is harder than copper
 - aluminium is less soluble in water than copper.
- A coil of copper wire can be made by pulling a larger piece of copper metal. This is due to which of the following properties of copper?
 - Malleability
 - Ductility
 - Tensile strength
 - Conductivity

(Page 44) **U**
- A copper coin is kept immersed in a solution of silver nitrate for some time. What will happen to the coin and the colour of the solution?
 - Silver metal will be deposited on coin and solution will turn blue.
 - Solution will remain colourless and coin will turn blue.
 - Both solution and the coin will turn blue.
 - Both solution and the coin will become colourless.

Ap
- A metal dissolved in mercury is called a/an
 - suspension
 - emulsion
 - amalgam
 - solution.

R
- When a metal X is added to dilute HCl solution, there is no evolution of gas. The metal X is
 - K
 - Na
 - Ag
 - Zn

U
- The correct order of increasing chemical reactivity is
 - $Zn < Fe < Mg < K$
 - $Fe < Mg < Zn < K$
 - $Fe < Mg < K < Zn$
 - $Fe < Zn < Mg < K$

R
- Which of the following elements produces basic oxide on reacting with oxygen?
 - Chlorine
 - Sulphur
 - Phosphorus
 - Magnesium

(Page 51)
- Tungsten is used in electric bulbs because
 - it is sonorous
 - it has high melting point
 - it has high tensile strength
 - it has high density.

15. Which of the following metal does not react with water?
(a) Na (b) Mg (c) Cu (d) Fe (Page 51)

16. Identify the most reactive metal.
(a) Iron (b) Gold
(c) Zinc (d) Potassium

17. An element X forms an oxide XO which turns red litmus blue. Identify X.
(a) A metal (b) A non-metal
(c) A metalloids (d) Any of these (Page 51) **U**

18. When steam is passed through zinc then
(a) zinc oxide is formed
(b) zinc hydroxide is formed
(c) hydrochloric acid is formed
(d) water is formed. **R**

19. Metal which does not react with steam is
(a) potassium (b) iron
(c) magnesium (d) silver.

Corrosion, Prevention of Corrosion and Alloys

20. An alloy used in making bodies of aircraft is
(a) duralumin (b) steel
(c) zinc-amalgam (d) solder. **R**

21. The metals are eaten away when they are exposed to water, oxygen and other chemicals. What is this process known as?

- (a) Vulcanization (b) Corrosion
(c) Galvanisation (d) Amalgamation

22. Bronze is an example of

- (a) an element
(b) a compound of copper and zinc
(c) a mixture of copper and lead
(d) a mixture of copper and tin. (Page 54) **R**

23. The metal which is present in brass, bronze and German silver is

- (a) Mg (b) Zn (c) Cu (d) Al **R**

24. The metal that forms a self protecting film of oxide to prevent corrosion is

- (a) Cu (b) Al (c) Pt (d) Au

25. Two metals are melted and mixed together. The resulting mass is cooled to solidify. What is this solid called?

- (a) An alloy (b) A non-metal
(c) An oxide (d) A mineral (Page 54)

26. Which of the following methods is suitable for preventing an iron frying pan from rusting?

- (a) Applying grease
(b) Applying paint
(c) Applying a coating of zinc
(d) All of the above.

Non-metals and their Physical and Chemical Properties

27. Which of these elements is used as an antiseptic in medicine?

- (a) Carbon (b) Oxygen
(c) Nitrogen (d) Iodine (Page 54) **R**

28. Which of the following does not contain a metallic element?

- (a) Sodium chloride (b) Silicon dioxide
(c) Magnesium oxide (d) Calcium hydroxide **U**

29. The non-metal used in vulcanization of rubber is

- (a) sulphur (b) carbon
(c) phosphorus (d) nitrogen. **R**

30. Which of the following is a non-metal?

- (a) Tungsten (b) Mercury
(c) Graphite (d) Platinum (Page 54) **U**

31. Which of the following is not a property of a non-metal?

- (a) Forms anions (b) Forms acidic oxides
(c) High tensile strength (d) Low density

32. Which of the following is not a basic oxide?

- (a) Sodium oxide (b) Calcium oxide
(c) Nitrogen dioxide (d) Iron oxide **U**

LEVEL - 2

33. An unknown metal X when placed in copper sulphate solution gives a red brown deposit. When placed in magnesium sulphate solution, gives no reaction. Identify element X.

- (a) Sodium (b) Potassium
(c) Silver (d) Iron

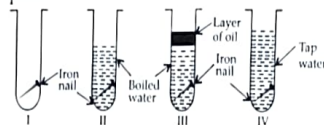
34. Which of the following metals on reaction with sodium hydroxide solution produce hydrogen gas?

1. Cu 2. Al
3. Fe 4. Zn

- (a) 2 and 3 (b) 2 and 4
(c) 1 and 4 (d) 2 only

35. Which of the following is not a correct match?
(P) Sodium - It is so soft that it can be cut with a knife
(Q) Phosphorus - Stored in water
(R) Copper - Acquires a dull green coating on exposure to air
(S) Charcoal - On burning produces a basic gas
(a) P (b) Q (c) R (d) S

36. In which test tubes, the rusting of iron nail will take place?



- (a) I and IV (b) I, II and IV
(c) II and III (d) II, III and IV **U**

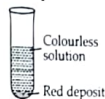
37. A highly reactive element X is stored under water. It readily reacts with oxygen of air to give a compound Y which dissolves in water. The aqueous solution of Y changes blue litmus solution to red. The element X is

- (a) sodium (b) sulphur
(c) phosphorus (d) potassium. **An**

38. Two elements A and B on burning in air give corresponding oxides. Oxides of both A and B are soluble in water. The aqueous solution of oxide of A is alkaline and reacts with aqueous solution of oxide of B to give another compound. Identify A and B.

- (a) A and B both are metals.
(b) A and B both are non-metals.
(c) A is metal and B is non-metal.
(d) A is non-metal and B is metal. **An**

39. When few granules of sample X are added to a solution of copper sulphate, the changes observed are shown in the figure. Identify sample X and red deposit.



- (a) Fe, Zn (b) Zn, Cu
(c) Cu, Zn (d) Fe, Cu

40. Choose the reactions which are not feasible
I. Iron + Zinc sulphate → Iron sulphate + Zinc
II. Magnesium + Silver nitrate → Magnesium nitrate + Silver
III. Copper + dil. Sulphuric acid → Copper sulphate + Hydrogen
IV. Zinc + Iron sulphate → Iron + Zinc sulphate
(a) I, II and III (b) III and IV only
(c) I and III only (d) I, II, III and IV **Ap**

41. Neha sets up an electric circuit as shown in the figure by using copper wire. She repeated the experiment with:

- I → Aluminium foil, II → Iron nail, III → Coal, IV → Graphite

In which cases the bulb will light up?

- (a) I and II only (b) III and IV only
(c) I, II and IV (d) I, II, III and IV

42. Study the table carefully and select the appropriate option.

Sample	Conductor of electricity	Malleability	Lustrous
W	✓	✓	✓
X	✓	×	✓
Y	✓	×	✓
Z	×	×	✓

W X Y Z

- (a) Potassium Sodium Graphite Aluminium
(b) Graphite Aluminium Sodium Potassium
(c) Sodium Aluminium Potassium Graphite
(d) Aluminium Sodium Graphite Iodine

43. Some materials like magnesium ribbon, aluminium foil, copper wire and charcoal (powder) were taken in different test tubes labelled as P, Q, R and S. 5 mL of dilute hydrochloric acid was added to each test tube. When a burning matchstick is brought near the mouth of each test tube, in which cases pop sound would be heard?

- (a) Only P (b) Only P and Q
(c) Only Q and S (d) Only P, Q and R

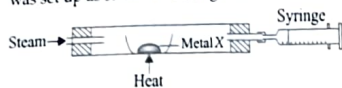
44. When a copper vessel is exposed to moist air for long, it acquires a dull green coating. The reaction can be represented as
- $$2\text{Cu} + \text{W} + \text{X} + \text{Y} \rightarrow \text{Cu}(\text{OH})_2 + \text{Z}$$
- moist air

W, X, Y and Z are respectively

- (a) N_2 , CO_2 , O_2 and CuO
 (b) H_2O , N_2 , CO_2 and O_2
 (c) H_2O , CO_2 , O_2 and CuCO_3
 (d) H_2O , CO_2 , O_2 and CuO

[R]

45. Ms. Komal, a science teacher demonstrated an activity in the science lab and the experiment was set up as shown in the figure.



The metal 'X' reacts with steam to form a gas 'Y' which is collected in the syringe.

Which of the following best describes the metal 'X' and gas 'Y'?

- (a) Metal 'X' is silver and gas 'Y' is hydrogen which burns with a pop sound.
 (b) Metal 'X' is sodium and gas 'Y' is carbon dioxide which gives white precipitate with lime water.
 (c) Metal 'X' is iron and gas 'Y' is hydrogen which burns with a pop sound.
 (d) Metal 'X' is magnesium and gas 'Y' is carbon dioxide which gives white precipitate with lime water.

[Cr]

LEVEL - 3 (HOTS)

Competency Focused Questions (CFQs)

46. A copper vessel exposed to moist air for a long time is shown in figure I and sulphur powder is burnt in oxygen as shown in figure II.

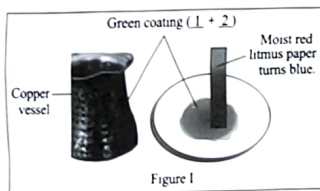


Figure I

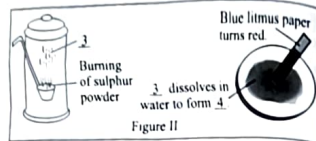


Figure II

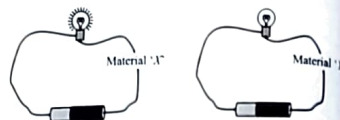
1, 2, 3 and 4 are respectively

- (a) CuO , CO_2 , SO_2 , H_2SO_3
 (b) CuCO_3 , $\text{Cu}(\text{OH})_2$, H_2SO_3 , SO_2
 (c) Cu , CuO , H_2S , H_2SO_4
 (d) $\text{Cu}(\text{OH})_2$, CuCO_3 , SO_2 , H_2SO_3

[An]

47. Rashmi has performed an activity to test whether the following materials conduct electricity or not by putting them in the circuit, one at a time as shown in the figure.

Materials			
Aluminium foil (1)	A piece of stone (2)	Copper wire (3)	
Graphite (4)	Sulphur (5)	Silver coin (6)	

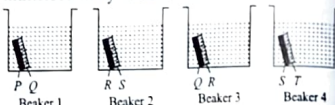


Materials X and Y could be

- X
 (a) 1, 3, 4, 6
 (b) 1, 5, 6
 (c) 1, 3, 6
 (d) 2, 3, 4, 5
- Y
 (a) 2, 5
 (b) 2, 3, 4
 (c) 2, 4, 5
 (d) 1, 6

[An]

48. Each of the given beakers contains two metal strips of same size fastened together and immersed in hydrochloric acid.



After 5 minutes, amount of Q ions formed is greater than P ions in beaker 1, S ions is greater than R ions in beaker 2, R ions is greater than Q ions in beaker 3, and T ions is greater than S ions in beaker 4.

Correct order of reactivity of these metals is

- (a) $P > Q > R > S > T$ (b) $R < P < T < S < Q$
 (c) $T > R > Q > P > S$ (d) $P < Q < R < S < T$

[Cr]

49. Read the table carefully.

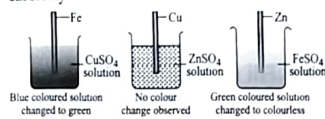
S. No.	Reaction	Product formed	Characteristic test
1.	Copper vessel exposed to moist air	Green coating	Turns red litmus solution blue
2.	Aluminium foil dipped in fresh solution of sodium hydroxide	Colourless, odourless gas	Burns with a pop sound
3.	Rusting of iron	Reddish brown deposit	Turns red litmus solution blue
4.	Burning of sulphur powder	Colourless, suffocating gas	Turns blue litmus solution red

Identify the products formed and their nature.

- (a) (1) $\text{Cu}(\text{OH})_2$; CuCO_3 ; basic
 (2) H_2 ; neutral
 (3) Fe_2O_3 ; basic
 (4) SO_2 ; acidic
 (b) (1) CuO ; basic
 (2) O_2 ; neutral
 (3) Fe_2O_3 ; basic
 (4) H_2S ; acidic
 (c) (1) $\text{Cu}(\text{OH})_2$; CuCO_3 ; basic
 (2) H_2 ; neutral
 (3) Fe_2O_3 ; acidic
 (4) SO_2 ; basic
 (d) (1) Cu ; basic
 (2) H_2O ; neutral
 (3) FeO ; basic
 (4) SO_3 ; basic

[U]

50. Gitika, a class VII student arranged the following experimental set-up and observed the changes carefully.



On the basis of her observations identify the correct order of reactivity.

- (a) $\text{Zn} > \text{Cu} > \text{Fe}$ (b) $\text{Zn} > \text{Fe} > \text{Cu}$
 (c) $\text{Fe} > \text{Zn} > \text{Cu}$ (d) $\text{Cu} > \text{Fe} > \text{Zn}$

[Cr]

51. Niharika categorised the different elements as shown below:

	Characteristic	Na	Al	S	Fe	Cu	P
(i)	Metal	✓	✓	✗	✓	✓	✗
(ii)	Hardness	✓	✗	✗	✓	✓	✗
(iii)	Malleability	✗	✗	✗	✗	✓	✓
(iv)	Ductility	✗	✗	✓	✓	✓	✗
(v)	Conductor of electricity	✓	✓	✗	✓	✓	✗

Which of the characteristics is not correctly matched by Niharika?

- (a) (i), (iii) and (v)
 (b) (i), (ii), (iv) and (v)
 (c) (ii), (iii) and (iv)
 (d) (i), (ii), (iii), (iv) and (v)

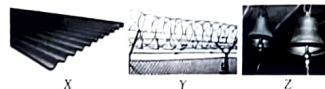
[An]

52. Any article of iron if left in open for some time, acquires a film of brownish substance (rust) having chemical formula _____.

- (a) Fe_2O_4 (b) $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
 (c) FeO (d) $\text{Fe}(\text{OH})_2$

[R]

53. Few uses of metals are shown below.

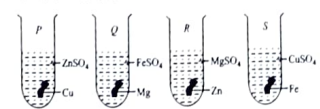


Identify the properties which are responsible for the uses X, Y and Z respectively.

- (a) Conductivity, Malleability, Ductility
 (b) Ductility, Malleability, Sonorous
 (c) Sonorous, Malleability, Ductility
 (d) Malleability, Ductility, Sonorous

[U]

54. A student performed the first set of experiments with four test tubes containing solutions and metal pieces as indicated in the figure. In the second set of experiments, the metal pieces are interchanged between test tubes P and Q and also between test tubes R and S.

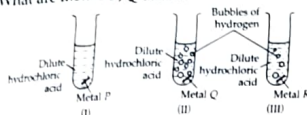


In which test tube(s), no reaction will occur in both sets of experiments?

- (a) P and Q (b) R
 (c) S (d) R and S

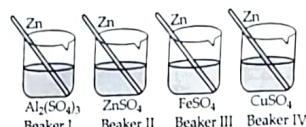
[An]

55. The given diagrams show the reactions of three metals with dilute hydrochloric acid. What are metals P, Q and R?



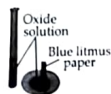
- | P | Q | R |
|---------------|-----------|-----------|
| (a) Copper | Magnesium | Zinc |
| (b) Copper | Zinc | Magnesium |
| (c) Magnesium | Zinc | Copper |
| (d) Zinc | Magnesium | Copper |

two hours, she noted the colour of each solution again. Mark the beakers in which colour change is observed?



- (a) I and II
(b) II and III
(c) III and IV
(d) I and IV

59. Samples of four different oxides were taken and dissolved in water separately to form the respective oxide solution. The four solutions were then tested for their acidic/basic nature. When dissolved in water, which of the following oxides, is likely to turn blue litmus red?



- (a) SO_2
(b) MgO
(c) Fe_2O_3
(d) CaO

60. Mona tries an experiment with four different metals namely sodium, magnesium, zinc, and copper taken in four different test tubes respectively, and added 5 mL of water to each test tube.



Mona observed that hydrogen gas is evolved in test tube 3, only when it was boiled. Test tube 3 contains only ____.

- (a) Zn
(b) Na
(c) Mg
(d) Cu

Fill in the Blanks

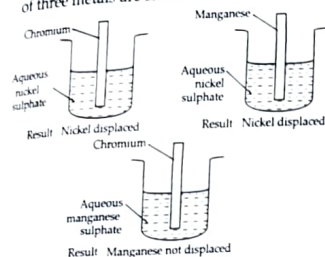
- Some metals like ____ are so soft that they can be even cut with knife.
- Metals have a tendency to ____ electrons to form positive ions.

- Unreactive metals, such as ____ and ____ can exist in uncombined state in nature.
- Copper and unreactive metals below it in the reactivity series do not react with dilute hydrochloric acid.
- A less reactive metal is able to displace a more reactive metal from its aqueous salt solution.
- Rust is hydrated iron(III) oxide i.e., $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$.
- Non-metals are soft and brittle. The only exception is graphite, an allotropic form of carbon, which is a non-metal but is the hardest natural substance known.
- Magnesium ribbon burns with dazzling pink light to form magnesium oxide.
- When iron rod is dipped in zinc sulphate solution, slowly colour of the solution turns light green and weight of iron rod keep on decreasing.
- Green layer deposited on copper on exposure to air is due to the formation of basic copper carbonate layer.
- Melting and boiling points of metals are very high except mercury and gallium.
- Copper does not burn in air even on strong heating. However, on heating in air for a long time, it is covered with a layer of green coloured copper(I) oxide.
- Lemon pickle is stored in aluminium vessel to keep it fresh for long time.
- Coal can be drawn into wires.
- Phosphorus is used in the vulcanization of rubber.
- Copper cannot displace silver from its salt solution but can displace iron.
- Nitrogen is used as preservative due to its inert nature. Some compounds of nitrogen are used as explosives as well.

True or False

- Steel is a mixture of elements but made up of mostly iron.

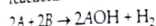
58. Three experiments to investigate the reactivities of three metals are shown as:



What is the correct order of reactivity (most reactive → least reactive) for these three metals?

- (a) Chromium, Manganese, Nickel
(b) Manganese, Chromium, Nickel
(c) Manganese, Nickel, Chromium
(d) Nickel, Chromium, Manganese

57. Identify A, B, C and D in the given sequence of reactions.



- | A | B | C | D |
|--------|----------------------|---------------|-------------------------|
| (a) Na | H_2O | SO_2 | H_2SO_3 |
| (b) Na | H_2O | SO_2 | H_2SO_4 |
| (c) K | H_2O | SO_2 | H_2SO_4 |
| (d) K | H_2O | SO_3 | H_2SO_3 |

58. Rutika noted the initial colour of the solutions in beakers I, II, III and IV. After inserting zinc rods in each solution and leaving undisturbed for

Match the Following

In this section, each question has two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (a), (b), (c) and (d) out of which one is correct.

- | List-I | List-II |
|------------------------|------------------------|
| (P) Aircrafts | 1. Stainless steel |
| (Q) Utensils | 2. Bronze |
| (R) Medals | 3. Magnalium |
| (S) Balance beam | 4. Duralumin |
| (a) P-3, Q-4, R-2, S-1 | (b) P-3, Q-1, R-2, S-4 |
| (c) P-4, Q-1, R-2, S-3 | (d) P-4, Q-3, R-1, S-2 |

- | List-I | List-II |
|------------------------|--------------------------|
| (P) Metallic oxide | 1. Turns blue litmus red |
| (Q) Phosphorus | 2. Stored in kerosene |
| (R) Non-metallic oxide | 3. Basic in nature |
| (S) Sodium | 4. Stored in water |
| (a) P-2, Q-3, R-4, S-1 | (b) P-4, Q-3, R-2, S-1 |
| (c) P-3, Q-4, R-2, S-1 | (d) P-3, Q-4, R-1, S-2 |

- | List-I | List-II |
|--|----------------|
| (P) Used in thermometers | (1) Iodine |
| (Q) Present in fertilisers | (2) Mercury |
| (R) Used to disinfect water | (3) Phosphorus |
| (S) Used as an antiseptic | (4) Chlorine |
| (a) (P)-(3), (Q)-(1), (R)-(2), (S)-(4) | |
| (b) (P)-(2), (Q)-(3), (R)-(4), (S)-(1) | |
| (c) (P)-(1), (Q)-(2), (R)-(3), (S)-(4) | |
| (d) (P)-(4), (Q)-(2), (R)-(1), (S)-(3) | |

- | List-I | List-II |
|---|------------------------|
| (P) The property of making resonating sound | 1. Ductility |
| (Q) The property to be drawn into wires | 2. Malleability |
| (R) The property to be beaten into sheets | 3. Sonority |
| (S) The property to withstand the longitudinal pull | 4. Tensile strength |
| (a) P-4, Q-1, R-2, S-3 | (b) P-3, Q-4, R-1, S-2 |
| (c) P-2, Q-3, R-4, S-1 | (d) P-3, Q-1, R-2, S-4 |

- | List-I | List-II |
|-----------------|--------------------------|
| (P) Zinc | 1. Liquid metal |
| (Q) Noble metal | 2. Liquid Non-metal |
| (R) Bromine | 3. Gold |
| (S) Mercury | 4. Used in Galvanisation |

- (a) P-3, Q-2, R-4, S-1 (b) P-4, Q-3, R-2, S-1
(c) P-4, Q-2, R-1, S-3 (d) P-2, Q-4, R-3, S-1

Assertion & Reason Type

Directions: In the following questions, a statement of assertion is followed by a statement of reason. Mark the correct choice as:

- (a) If both assertion and reason are true and reason is the correct explanation of assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of assertion.
(c) If assertion is true but reason is false.
(d) If both assertion and reason are false.

- Assertion** : Iron is found in free state in nature.
Reason : Iron is not highly reactive metal.
- Assertion** : Zinc is used in galvanisation of iron.
Reason : Its coating on iron articles increases their life by protecting them from rusting.
- Assertion** : Copper cannot displace hydrogen from acids.
Reason : It lies above hydrogen in the reactivity series.
- Assertion** : Aluminium appears dull in colour and does not react with water, acids or alkalies.
Reason : It has a protective layer of oxide on it.
- Assertion** : If a piece of copper is placed in silver nitrate solution, the solution becomes blue.
Reason : Displacement reaction takes place and copper goes into the solution.
- Assertion** : On beating with the help of a hammer, coal is converted into small pieces and finally into a powder.
Reason : Coal is made up of carbon which is malleable in nature.
- Assertion** : Most of the metals can be beaten into sheets.
Reason : Most of the metals are ductile.
- Assertion** : Graphite is a non-metal.
Reason : It is the only non-metal which is good conductor of electricity.

9. **Assertion** : A solution of carbon dioxide in water turns red litmus blue.

Reason : Carbon dioxide is basic oxide.

10. **Assertion** : Copper and aluminium are used to make electric wires.

Reason : Copper and aluminium are good conductors of electricity.

Comprehension Type

PASSAGE-I : The decay of metals by atmospheric oxygen and moisture is called corrosion. Few metals do not react with atmospheric gases and water and do not undergo corrosion, while others develop a dull layer on them or a coloured layer to spoil their shiny appearance or turn black or converted into powder. Few elements, on the other hand are protected due to presence of this dull layer of oxide on them.

- Aluminium does not react readily with air or water because
 - it occupies high position in electrochemical series
 - it lies below hydrogen in electrochemical series
 - it is covered with a layer of oxide which does not rub off
 - it is a noble metal.
- Iron reacts with air in presence of water and forms a brown powder which is called rust. Chemically rust is
 - iron oxide
 - hydrated iron oxide
 - iron sulphate
 - iron carbonate.
- Silver metal on exposure to air for a long time becomes black in colour. What is the reason behind it?
 - Silver reacts with sulphur present in air and forms silver sulphide.
 - Silver reacts with carbon dioxide present in air to form silver carbonate.
 - Silver reacts with water to form silver hydroxide.
 - Silver reacts with air to form silver oxide.
- The green layer developed on copper surface on exposure to air is due to
 - copper sulphate layer
 - basic copper carbonate layer
 - copper sulphate layer
 - copper nitrate layer.

PASSAGE-II : Certain metals have the capacity to displace some metals from their salt solutions. These reactions are known as metal displacement reactions. A metal placed higher in activity series can displace the metal occupying lower position from aqueous solution of its salt or a more reactive metal can displace less reactive metal from its salt solution.

- If a zinc rod is dipped in copper sulphate solution, following changes will be noticed. Mark the correct observations.
 - Zinc rod slowly loses its weight.
 - Copper settles at the bottom of the beaker.
 - Blue colour of copper sulphate disappears.

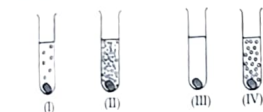
(a) (i) and (ii) only (b) (i), (ii), and (iii)
(c) (i) and (iii) only (d) (iii) only
- Take aqueous solution of copper sulphate in one test tube and ferrous sulphate in another test tube. Dip an iron nail in copper sulphate and copper wire in ferrous sulphate solution. Mark the correct observation.
 - Blue colour in first test tube changes to light green and green in the second test tube changes to blue.
 - There is no reaction in both the test tubes.
 - Blue colour in first test tube changes to light green and no change in second test tube.
 - Blue colour remains as such and green colour in second test tube changes to blue.
- The above observations show that
 - copper is present above iron in the reactivity series
 - iron is present above copper in the reactivity series
 - both iron and copper are present above hydrogen in the reactivity series
 - reactivity series does not decide reactivity of the metals.

PASSAGE-III : Metals and non-metals react with oxygen to give oxides at different rates. The oxides formed by metals are basic in nature while oxides formed by non-metals are acidic in nature. The nature of oxides can be determined by testing the aqueous solution of oxide with litmus paper.

- The oxides of non-metals dissolve in water to give
 - alkalis
 - acids
 - carbonates
 - sulphates.
- Sulphur is burnt in air to give sulphur dioxide. It is dissolved in water and tested with litmus paper. Mark the correct observation.
 - Red litmus paper turns blue.
 - Blue litmus paper turns red.
 - There is no change in the litmus paper.
 - Red litmus paper changes to green.
- Magnesium ribbon on burning in air gives a white powder which when dissolved in water turns red litmus blue. The reason for this change is that
 - MgO is a basic oxide
 - MgO is an acidic oxide
 - MgO is a very reactive oxide
 - MgO is not a reactive oxide

PASSAGE-IV : Metals in general have a tendency to evolve hydrogen gas on reacting with acids. In the activity series the metals that are placed above hydrogen evolve the gas on reaction with acids while the metals placed below hydrogen do not evolve the gas on reaction with acids. These metals are regarded as inactive metals. Metals placed at the bottom of the series are called noble metals.

- Which metal would not produce bubbles of hydrogen gas when added to dilute hydrochloric acid?
 - Magnesium
 - Sodium
 - Iron
 - Silver
- Pieces of copper, lead, aluminium and zinc are added to dilute hydrochloric acid. Which of the following test tubes most likely contains zinc?



(a) (I) (b) (II) (c) (III) (d) (IV)

- Silver and gold do not react with oxygen even at very high temperature while copper reacts on prolonged heating. The reactivity of these metals is
 - $\text{Cu} > \text{Ag} > \text{Au}$
 - $\text{Cu} = \text{Ag} > \text{Au}$
 - $\text{Cu} = \text{Ag} = \text{Au}$
 - $\text{Au} > \text{Cu} > \text{Ag}$

Subjective Problems

Very Short Answer Type

- What are the elements which show properties of both metals and non-metals called?
- Name the reaction in which more reactive metal displaces less reactive metal from its salt solution.
- What is galvanization?
- Why aluminium and copper metals are used for making utensils?
- What is rust chemically known as?
- What happens when a solution of metal oxide is tested with (i) blue litmus and (ii) red litmus?
- Name the alloy of copper which is used in making utensils and vessels.
- List four physical properties of metals.
- Name the alloy of lead used in joining metals for electrical work.
- Why are metals good conductors of electricity?
- Name two most malleable metals.
- Arrange Ca, Zn, Fe, Au, Ag and Cu in order of decreasing reactivity.
- What is the most common property of the metals lying at the bottom of the reactivity series?
- What is an amalgam?
- In what respect does graphite resemble a metal?

Short Answer Type

- Complete the given table about properties of metals.

Property	Meaning
(a)	Can be beaten into thin sheets
(b)	Can be pulled into wires
(c)	Gives a ringing sound when hit
(d)	Has a shiny appearance

- For the following questions, choose answers from the list of elements - mercury, carbon, hydrogen, tin, sulphur, bromine, magnesium, silicon.

- Which of the elements are solids?
- Which of the elements are liquids?
- Which of the elements are gases?
- Which of the elements are metals?
- Which of the elements are non-metals?
- Which of the following elements are sonorous?

- What happens when a magnesium ribbon is heated in presence of air?
- List few important uses of metals.
- What will happen if you drop
 - some pieces of magnesium ribbon into blue copper sulphate solution?
 - some tin pieces into green ferrous sulphate solution?
- Gold, platinum and silver are used to make jewellery. Give reasons.
- Why do some metals acquire a dull appearance on exposure to air for a long time?
- Give reasons why metals are good conductors, whereas non-metals are bad conductors of electricity.
- Why are some metals light in weight? Give examples of few light metals.
- Arrange the metals Na, Ca and Mg in decreasing order of reactivity on the basis of reaction with water.
- How do alloys, brass and bronze differ in composition?
- Aluminium metal is not as reactive as expected. Why?
- Which of the following metals will give hydrogen with dilute hydrochloric acid?
Fe, Cu, Mg
- Explain why aluminium metal can displace copper from copper sulphate solution but copper cannot displace aluminium from aluminium sulphate solution?
- Compare the changes taking place in an iron piece and a wood log on beating with a hammer.

Long Answer Type

- Two properties of four substances A, B, C and D are given in the table.

Substance	Melting point (°C)	Electrical conductivity	
		Solid	Molten form
A	1100	Good	Good
B	1234	Poor	Good
C	2027	Poor	Poor
D	15	Poor	Poor

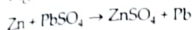
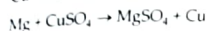
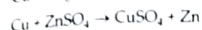
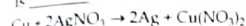
Which of these substances is most likely a metal and why?

- Identify the non-metals on the basis of their uses:
 - Non-metal essential for all living beings.
 - Non-metal used in fertilizers to enhance growth of plants.
 - Non-metal used in water purification process.
 - Non-metal used in purple coloured solution which is applied on wounds and used as an antiseptic.
 - Non-metal used in crackers.
- What are the harmful effects of corrosion? Mention a type of corrosion which is helpful to the metal.
- Define displacement reaction. An iron knife kept in blue copper sulphate solution turns the blue solution into light green. Explain.
- With a suitable activity show that sulphur burns in air to form a compound which is acidic in nature.

Integer/Numerical Value Type

- Number of metals which lie above hydrogen out of the following is _____.
Sodium, lead, copper, platinum, zinc, mercury
- In the reaction, $x\text{Na} + \text{O}_2 \rightarrow \frac{x}{2}\text{Na}_2\text{O}$, x is _____.
- Number of elements which form basic oxide is _____.
Magnesium, aluminium, carbon, sulphur, iron, potassium, zinc
- Number of elements which will not react even with boiling water is _____.
Sodium, iron, zinc, copper, magnesium, silver

5. Number of possible reactions out of the following is _____



Case Based Questions

Case I : Different materials are present around us. These materials can be divided into two categories, metals and non-metals. These two differ from each other in various physical and chemical properties, like metals are generally lustrous but non-metals are not. But few exceptions are also there.

1. The property of metals by which they can be beaten into thin sheets is called
- ductility
 - sonorosity
 - lustre
 - malleability.

2. Which of the following elements is soft in nature?
- Sodium
 - Aluminium
 - Copper
 - Iron

3. Some elements are given below:
- Sulphur
 - Oxygen
 - Phosphorus
 - Copper
 - Sodium

Which of the given elements are non-metals?

- (ii), (iii) and (iv)
 - (i), (ii) and (iii)
 - (i), (iv) and (v)
 - (i), (iii) and (v)
4. Class-7th students performed an activity in their class as shown in the figure.



They used several other materials listed below, in place of the iron nail :

Iron rod, Copper wire, Coal piece, Wood piece, Sulphur, Aluminium foil, A small stone

In which of the following groups (i) to (iv) bulb will not glow at all?

- Coal piece and sulphur
 - Iron rod, coal piece, small stone
 - Iron rod, copper wire, aluminium foil
 - Wood piece and small stone
- (ii) and (iii) only
 - (i) and (iii) only
 - (iv) only
 - (i) and (iv) only

5. Which of the following materials can be drawn into long wire?

- Aluminium, iron, gold
- Coal, sulphur, wood
- Coal, sulphur, aluminium
- Copper, carbon, aluminium

Case II : Metals are generally very reactive, they react with oxygen or moisture present in air and form oxides. These oxides when dissolve in water form hydroxides. Non-metals also react with oxygen to form oxides which when dissolve in water form acids.

1. When a copper vessel is exposed to moist air for long time, it acquires a dull green coating. The green material is a mixture of
- copper oxide
 - copper hydroxide
 - copper carbonate
 - copper sulphide

- I and II
- I and III
- II and III
- III and IV

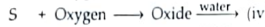
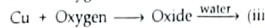
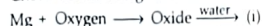
2. Sulphur is burnt in a glass tumbler and the tumbler was covered with a lid so that the gas produced does not escape. Now, a small quantity of water is added to the tumbler and the solution is checked with litmus paper. Identify the gas produced and the colour change in litmus paper.

- CO_2 , blue litmus to red
- CO_2 , red litmus to blue
- SO_2 , red litmus to blue
- SO_2 , blue litmus to red

3. Which of the following is not a basic oxide?

- Na_2O
- CO_2
- CaO
- MgO

4. Check the following reactions



Select the correct statement regarding acidic and basic nature of compounds (i) to (v)?

- (i) to (iii) are acidic while (iv) and (v) are basic.
- (i) to (iii) are basic while (iv) and (v) are acidic.
- (i), (ii) are basic while (iii) to (v) are acidic.
- (i), (ii) are acidic while (iii) to (v) are basic.

5. Phosphorus is kept in water because

- it is a very reactive metal and reacts explosively with air
- it is a noble metal and catches fire when reacts with air
- it is an inert non-metal and reacts with oxygen from air
- it is a very reactive non-metal and catches fire when reacts with oxygen from air.

Case III : Metals react with water but non-metals do not. Reactivity of metals may vary from metal to metal, few reacts vigorously while others not. Metals and non-metals also react with acids differently. Metals show different reactivity towards acid and base. Reactions of non-metals with bases are complex.

1. Name the gas evolved when a metal reacts with water and how the gas can be tested?
- Hydrogen, burns with a pop sound
 - Carbon dioxide, burns with a pop sound
 - Carbon dioxide, turns lime water milky
 - Hydrogen, turns lime water milky

2. The metals which react slowly even on reaction with steam are

- Mg and Cu
- Na and Cu
- Na and K
- Zn and Fe

3. How does metals and non-metals react with dilute acids?

- Both form water and salt.
- Both form water and oxygen.
- Metals form salt and hydrogen while non-metals do not react with dilute acids.
- Both do not react with dilute acids.

4. Which of the following is not correct about copper?

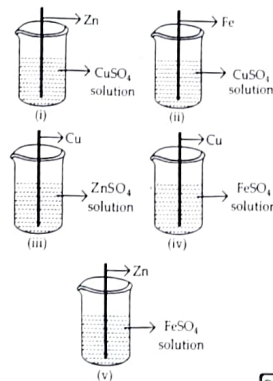
- Copper does not react with water.
- Copper does not react with dil. HCl.
- Copper reacts with moist air to form basic green coating.
- None of these

5. Which of the following is/are correct?

- Sodium + Water $\longrightarrow$ Sodium hydroxide + Hydrogen gas
 - Sulphur + Water $\longrightarrow$ No reaction
 - Zinc + Sodium hydroxide $\longrightarrow$ Sodium zincate + Hydrogen gas
- (i) and (ii) only
 - (i), (ii) and (iii)
 - (iii) only
 - (ii) only

Case IV : Few metals are more reactive than others. The metals which are more reactive can displace less reactive metal from its salt solution, but less reactive metals cannot displace more reactive metals from their salt solutions.

Observe the given figures carefully.



1. In which of the given beakers, reaction will not take place?

- (i), (ii) and (iii) only
- (iii), (iv) and (v) only
- (ii), (iii) and (iv) only
- (iii) and (iv) only

2. When a strip of zinc is placed in iron sulphate solution, then zinc metal displaces iron metal from iron sulphate solution this implies that
- zinc is less reactive than iron
 - iron is less reactive than zinc
 - both are equally reactive
 - it has nothing to do with reactivity.

3. A zinc rod is dipped in copper sulphate solution. Which of the following represent(s) the correct observation(s).

- Zinc rod slowly loses its weight.
 - Copper settles at the bottom of the beaker.
 - Blue colour of copper sulphate solution disappears.
- (i) and (ii) only
 - (i) and (iii) only
 - (iii) only
 - All observations are correct.

4. When a strip of iron metal is placed in a copper sulphate solution for sometime, then

- blue colour of CuSO_4 fades and reddish-brown coating of copper metal is deposited on the iron strip
- blue colour of CuSO_4 fades and grey coating of copper metal is deposited on iron strip
- no reaction will take place
- blue colour of copper sulphate changes to brownish.

Which of the following pairs will not give displacement reaction?

- Sodium chloride solution and copper metal
 - Magnesium chloride solution and aluminium metal
 - Iron sulphate solution and silver metal
 - Silver nitrate solution and copper metal
- (ii) and (iv) only
 - (i) and (iv) only
 - (i), (ii) and (iii) only
 - (iv) only

Case V : We are surrounded by several materials or matter which is present in the form of elements, compounds and mixtures. Most of the elements are metals and the remaining are non-metals or metalloids. The metals are very useful in our everyday life due to their properties like hardness, malleability, ductility and conductivity. Non-metals and metalloids are also very useful to us.

1. Which of the following non-metals is used in fertilizers to enhance the fertility of soil?

- Oxygen
- Nitrogen
- Chlorine
- Potassium

2. Which of the following metals is used for making both cooking utensils and electric wires?

- Magnesium
- Zinc
- Sodium
- Copper

3. Which of the following non-metals is used as an antiseptic?

- Iodine
- Chlorine
- Hydrogen
- Sulphur

Changes Around Us : Physical and Chemical

CHAPTER 03

Learning Outcomes...

- Physical and Chemical Changes : Characteristics and Differences
- Crystallisation
- Rusting of Iron and Its Prevention
- Combustion : Conditions and Types
- Fire Extinguishers
- Fuel – Classification and Calorific Value
- When Physical and Chemical Changes Occur Together
- Different Types of Change
- Weathering of Rocks
- Erosion

Introduction

Changes take place around us all the time. The changes differ in terms of appearance, colour, composition, formation of new products or any other change. The changes can be divided into different types depending upon the nature of the change taking place. Changes can be slow or fast, desirable or undesirable, periodic or non-periodic. Broadly, these changes are of two types – physical and chemical.

Physical Change

The change in which a substance undergoes a change in its physical properties are called **physical change**. Properties such as shape, size, colour and state of a substance are called **physical properties**.

Characteristics of Physical Changes

- No new substances are formed.
- The properties of the substance remains same even if the physical state changes.
- There may or may not be evolution of heat.
- Properties like shape, size, texture, colour, smell, etc. may change.
- Chemical properties of the substance do not change.
- In a physical change the things can generally get back to their original form after the change.

ACTIVITY CORNER



To study a physical change using a chalk

- » **Materials required :** Chalk, water
- » **Procedure :** Crush a chalk piece into powder. Add a little water to the chalk powder and make a paste. Roll it into the shape of the chalk again.